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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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D7.4 Data Management Plan

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1 Introduction

This is the first version (M6) of the SymAware Data Management Plan. From chapter 2 onwards the DMP follows the EU Grants Data Management Template (HE: v1.0). In this introduction chapter we give a brief high-level summary of the SymAware project as a reference.

A final version of the DMP will be delivered at the end of the project (D7.5, M36).

1.1 SymAware: High-level description of the project

SymAware addresses the fundamental need for a new conceptual framework for awareness in multi- agent systems (MASs) that is compatible with the internal models and specifications of robotic agents and that enables safe simultaneous operation of collaborating autonomous agents and hu- mans. The goal of SymAware is to provide a comprehensive framework for situational awareness to support sustainable autonomy via agents that actively perceive risks and collaborate with other robots and humans to improve their awareness and understanding while fulfilling complex and dynamically changing tasks. The SymAware framework will use compositional logic, symbolic computations, formal reasoning, and uncertainty quantification to characterize and support situational awareness of MAS in its various dimensions, sustaining awareness by learning in social contexts, quantifying risks based on limited knowledge, and formulating risk-aware negotiation of task distributions. These objectives will be achieved in SymAware through

1. logical characterization of awareness using symbolic methods,
2. quantifying the symbolic reasoning for awareness with spatial and temporal ingredients for decision making,
3. risk awareness via quantified knowledge,
4. quantifying and communicating knowledge awareness,
5. demonstrating awareness engineering in aviation and automotive use cases, and
6. identifying requirements for ethical and trustworthy awareness in human-agent interaction.

The objectives of SymAware address the “Awareness Inside” Challenge of the European innovation Council (EIC) by extending and formalizing human-based models of situational awareness and by providing a novel conceptual situational awareness framework for MASs that encompasses logical characterization and integrative formal reasoning of interdependent awareness dimensions including knowledge, spatiotemporal, risk and social dimensions. This will support transitioning to safe mixed operation of autonomous agents and humans.

2 Data Summary

All partners indicated their data summary answering the questions below in subsequent paragraphs.

1. Will you re-use any existing data and what will you re-use it for? State the reasons if re-use of any existing data has been considered but discarded.
2. What types and formats of data will the project generate or re-use?
3. What is the purpose of the data generation or re-use and its relation to the objectives of the project?
4. What is the expected size of the data that you intend to generate or re-use?
5. What is the origin/provenance of the data, either generated or re-used?
6. To whom might your data be useful ('data utility'), outside your project?

2.1 MPI-SWS

1. We will not use any existing data.
2. We will generate synthetic data from algorithms and computer simulations. We are performing simulations in Python and MATLAB.
3. The purpose of the data generation is to conduct research on situational and knowledge awareness in MASs that are operating in a dynamic and unstructured environment. These are also the key objectives related to WP1 and WP4 of the project.
4. The expected size of the data is in Gigabytes.
5. The data will be generated through computer simulations using different algorithms.
6. The data can be useful for the research communities and software companies.

2.2 UU

1. We will not be re-using any existing data.
2. We will be collecting quantitative and qualitative data from experiments with human participants. This will include (but not limited to): behavioral and task data from participants which could be collected via a computer or a tablet, video and audio recordings of participants, transcripts of focus group audio recordings, questionnaire data, sketching and design outputs.
3. The generated data from the human-agent and human-robot interaction (HAI/HRI) studies will be analyzed to explore how the human participants interact with the artificial agents, simulation environment, and robots in the studies. The qualitative data from the questionnaires and the participatory design studies will be analyzed to identify participants' opinions of the interaction, and in a collaborative way identify the ethical risks of the use case scenarios, as well as develop prototypes for the SymAware technologies.
4. The expected data size is in gigabytes.
5. Depending on the type of data, it would either be generated by the human participants directly (i.e. behavioral and task data, questionnaires and other participatory design data), or indirectly (audio and video recordings of the participants).
6. The analysis of the data (and the data itself in an anonymized form) could be useful for the HAI/HRI research communities, particularly for research groups attempting replication studies with a similar premise to ours.

2.3 NLR

1. We will not use existing data.
2. We will generate risk estimates for various scenarios in unmanned aircraft systems traffic management (UTM) using computer simulation.
3. The purpose of the data generation is research on the awareness framework in the aviation use case in WP5.
4. The expected size of the data is in Megabytes.
5. The risk estimates are generated using computer simulation.
6. The risk estimates for the scenarios will be reported in D5.3 and they will be useful for UTM research and developers.

2.4 KTH

1. We will not use existing data.
2. We will generate data by deriving mathematical algorithms, performing computer simulations using MATLAB, and conducting experiments using experimental robotic platforms.
3. The purpose of the data generation is to investigate the spatiotemporal and risk awareness in multi-agent systems task planning and control operations, which are in line with the research objectives in WP2 and WP3 of the project.
4. The expected size of the data is in Gigabytes.
5. The data will be generated using computer simulations and experiments through experimental robotic platforms.
6. The data could be useful for researchers in the scientific and robotics communities, and software developers.

2.5 TUE

1. We will not use existing data.
2. We will generate data by deriving mathematical algorithms, performing computer simulations using MATLAB, and conducting experiments using experimental robotic platforms.
3. The purpose of the data generation is to investigate the spatiotemporal and risk awareness in multi-agent systems task planning and control operations, which are in line with the research objectives in WP3 and WP5 of the project.
4. The expected size of the data is in Gigabytes.
5. The data will be generated using computer simulations and experiments through experimental robotic platforms.
6. The data could be useful for researchers in the scientific and robotics communities, and software developers.

2.6 SISW

1. We will not use existing data.
2. We will generate simulation experiments/data and office documents for documentation.

3. The purpose of the data generation is to enable testing and verification of activities in WP3 and WP5.
4. The expected size of the data is in Gigabytes
5. The data will be generated using computer simulations.
6. The data could be useful for researchers in the scientific and verification and validation community in the automotive domain.

3 FAIR data

3.1 Making data findable, including provisions for metadata

Data generated and identified by the research team to be shared will be deposited in a data repository, such as Zenodo. Zenodo (https://zenodo.org/) is a general-purpose open repository developed under the European OpenAIRE program and operated by CERN. Zenodo supports Open Science and in that context the FAIR data principles.
During the research, the metadata will use standard formats as much as possible, and a README file will be created.
The metadata associated with the data will be available through the same repository. The metadata schemes used by the trusted repository include appropriate keywords making the data sets belonging to publications findable. By making use of the metadata scheme provided by the by the zenodo.org, the metadata are indexed and searchable. The repositories support harvesting of metadata via a standard protocol. We follow the Dublin Core metadata schema for data description and aim to document all "relevant" data in such a way that they are understandable to researchers in the field.
For publication data and data sets deposited in open access repositories unique identifiers such as Digital Object Identifiers (DOIs) will be used. A digital object identifier (DOI) is automatically created for each dataset uploaded in Zenodo, which makes the stored items easily citable.

3.2 Making data accessible

All data underpinning research papers will be made available to other researchers at the time of the article's publication by depositing at the Zenodo repository, unless this is not possible due to patent applications/IP restriction. In that case some of the data may need to be embargoed. Additionally, intellectual property rights and ownership of the results are discussed in the Grant Agreement.
All raw and analysed data underlying a journal publication will be made available for verification/replication and reuse, i.e., will be made freely available together with the relevant metadata (including README.txt files) on the zenodo.org repository fulfilling the FAIR requirements.
As a lot of different Parties are involved in the project, not all of them have agreements with Zenodo. However, Zenodo is an all- purpose open data repository and the only limitation currently is related to the volume. A maximum volume is 50 GB per data set and we do not expect to exceed this. However, higher quotas can be requested and granted on a case-by-case basis. A digital object identifier (DOI) is automatically created for each dataset uploaded in Zenodo, which makes the stored items easily citable.
Since we are aiming to use GitHub for our software, from within Zenodo, we are aiming to **setup a webhook** in the GitHub repository which is triggered when we will make a new release at GitHub. Metadata is always openly available even for data under an embargo. The datasets that are deposited into the trusted repositories will be associated with one of the CC-BY Creative Commons licenses to protect the authorship of the data. Data and metadata will remain available for at least 20 years at zenodo.org.

3.3 Making data interoperable

The research data together with the required metadata belonging to scientific publications are stored in open standard file formats. We will use the metadata standard of the zenodo.org repository which is similar to Dublin Core. This facilitates the exchange and re-use of data inside and outside the consortium. Furthermore, the interoperability of the data and metadata is insured by making use of common vocabularies used in the research field literature. If necessary, new or uncommon vocabularies necessitating explanations, i.e. for interdisciplinary interoperability, will be clarified. Full text will be used before using any abbreviations.

In this project we will develop code (Python) and simulation models (MATLAB/Simulink). When stored in the trusted repository and GitHub repositories, clear instructions on how to use the code and models will be available through README files. Also, open-source code will include comments to make it readable. The code and models will be used in Workshops with the community of workshop followers and also for that purpose interoperability and re-use are a must.

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3.4 Increase data re-use

All the data is accompanied by rich metadata and documentation describing all steps in the chain from collection to final analysis. This is done by means of README files describing the used methodology, simulation tools, parameters, settings, analysis steps. Detailed documentation enables that all steps in the chain can be verified by others, ensuring the quality of the work.
The datasets that are deposited into the trusted repositories will be associated with one of the CC-BY Creative Commons licenses to protect the authorship of the data. The Attribution license CC-BY allows re-distribution and re-use of a licensed work on the condition that the original creator is appropriately credited, so the aim is to use this license. The exact license will be chosen when a dataset is finalized, and specifications on usage of the data are well established.
The data will remain openly accessible in the chosen open-access repository for at least 20 years for zenodo.org. For a throughout data description, we follow the Dublin Core metadata schema and README files.

4 Other Research Outputs

4.1 Introduction

In addition to the data generation and collection as described in the previous chapter, the project will also result in

1) Generic concepts and methodologies published in scientific publications,
2) Software implementations directly related to the published concepts and methodologies,
3) Use case specific models, interfaces and simulation results.

Items 1) and 2) will not be protected but published and shared open access and open source. Item 3), the results of the work on Use Cases in WP5 are expected to be largely open access as well but might contain elements that partners find valuable to protect as they might have business potential.

4.2 Scientific publications

Partners will publish scientific articles with Green or Gold-standard open access to share results generated from the project. Research data underlying scientific publications will be made available via open access data repositories as describe in Chapter 3. Each data set deposited is assigned a DOI so that it is uniquely identifiable and can be cited in scholarly articles.

4.3 Software implementations

Software implementations of the developed methodologies will be made available open access and open source on GitHub (using the BSD 3-clause license).
All related scientific publications will contain the link to the specific GitHub repo through the Zenodo GitHub integration, while vice versa the GitHub repository will refer back to the related open access scientific publications.

5 Allocation Of Resources

No specific resources have been allocated to making data FAIR. However, TU/e has allocated resources to support researcher with research data management, making data FAIR, open access publishing and any ethical and privacy issues. Moreover, the chosen open access data repository (e.g. Zenodo or even 4TU.ResearchData) will be free of charge for educational and informational use. At the moment, we don't expect any additional costs to be needed in regards to research data management and making data FAIR.

Every researcher involved in the project is accountable for responsible data management. However, the main applicant is primarily responsible for the data management and will allocate part of his time resources to ensure FAIR data management.

As Zenodo is a trusted data repository, it claims that items will be retained for the lifetime of the repository. This is currently the lifetime of the host laboratory CERN, which currently has an experimental programme defined for the next 20 years at least.

6 Data Security

During the project data will be stored at an institutional cloud storage solution (**ResearchDrive**) that is an ICT service of TU/e and hosted and backed up on the servers in the Netherlands. In case any (secondary) processing of personal data is involved, a suitable encryption will be used. The ResearchDrive cloud storage solution is accessible by all project partners.

All personal data that might be gathered during the research will be pseudonymized and available only to the researchers and stakeholders involved in the project. Only pseudonymized and aggregated data will be available for publishing and in the chosen data repository. Open access to research data under the principle "as open as possible, as closed as necessary" will be used when finalizing the scientific outcomes that will result from the processing of personal data.

After the project has ended, all scientific outcomes that will not be a subject to IP rights, or have any privacy and security issues, will be stored in a trusted Data Repository.

7 Ethics

While we do have a work package on Ethics - WP9 - the ethics deliverables D9.1 and D9.2 outline the strategies to identify and mitigate ethical risks concerned to the use of AI in our research. The only relevant ethics issues related to data collection and sharing involve the data from **human participants.**

The project will collect human data for feedback on Use Cases and developed software framework. We distinguish two types of human data collection as further described in the paragraphs below.

An external ethics advisor has been appointed. He will ensure that ethics issues are handled correctly.

7.1 Human data related to project Use Cases

During the project several Use Cases will be evaluated. This also includes obtaining human responses on the behavior of autonomous agents in these specific Use Cases. For these dedicated questionnaires will be developed and used. This type of human data collection will be covered by Uppsala University.

The research conducted at Uppsala University will be covered by an Ethical Approval which among other things contains a detailed indication of any possible ethical risks related to data collection and sharing, and our strategies to mitigate those risks.
In case of secondary processing of personal data by other partners, depending on the need of the research, the data transfer will be covered with a Data Transfer Agreement. Additionally, if applicable, the secondary processing of the personal data will be submitted for approval from the Ethical Review Board by the receiving partner, and also a Privacy Impact Assessment will be performed.

7.2 Human data related to using the SymAware software framework

The consortium will actively seek contact with parties interested in the project research. The aim is to build a **community of followers** that will be updated on project research progress but also invited to participate in workshops using the project developed software framework. To reach out to this community during project execution contact information (email addresses) needs to be obtained. We will disseminate the option to become a project follower using various communication channels, providing a link to a web-based registration tool – Copernica -where people can register providing us with their email address.

During registration of the followers, we will collect no more data than necessary for having good quality feedback. The registration details will be gathered and used only by TUE. Copernica adheres to Gereral Data Protection Regulations.

When people decide to participate in the workshops on the SymAware software framework we will also collect their feedback on using the software after the workshop. Collecting the feedback will be done by TUE. The feedback will available to other partners as well.

The personal data that will be collected in order to complete the recruitment process, will be deleted right after the process is over. As opinions might be considered personal data, and this might be needed

in order to get feedback from the participants on using the software, all such data will be pseudonymized and aggregated for scientific conclusions. Before the personal data is collected and processed, Informed Consent will be sent to potential participants, explaining the purpose of the research, the input needed from the participants, and the type of data that will be collected and processed. Additionally, the Informed Consent will serve to explain the participants that the participation is voluntary, and that their consent can be withdrawn. The research team will also seek approval from the Ethical Review Board of the institution. Due to collection of personal data during the research period, additionally Privacy Impact Assessment will be performed, to assess and mitigate the privacy risks of the data subjects.

8 Other issues

8.1 MPI-SWS

MPI-SWS is not using any national or departmental procedures for data management.

8.2 UU

We at Uppsala University will comply with the EU's data protection regulation (General Data Protection Regulation) and **handle data according to national Swedish law (The Archives Act)**. All data is stored without information that can identify participants. Participants will be assigned a random number, i.e. a unique code (P-ID) (eg in notes or file names we refer to a participant by a number, we record the number on the consent form so we can find relevant information if necessary). The code key is saved electronically on a locked computer at Uppsala University.

The code key and all data collected from the studies are treated confidentially and stored anonymously and securely in a password-protected computer at Uppsala University and are available on request to researchers in the SymAware project, who follow the same basic protocol for data storage. The computers will be accessible from Uppsala University's network.

Uppsala University is responsible for storing and managing data. Participants taking part in the study may be videotaped. We may display video recordings of these studies in scientific articles, conferences and events, but participants can opt out by ticking the box on the consent form (see attachment). Data collected during this study, e.g. data automatically extracted from the video recordings (e.g. the position of the main points of the face, such as the corners of the mouth or eyebrows) can be shared with peers and collaborators outside the project and displayed in publications anonymized and in numerical format (i.e. no video with identifiable information will shared but only numbers representing facial features and expressions) to facilitate transparency and openness in science.

Collected data will be processed for as long as necessary to ensure the quality of the research. Data is stored for the entire duration of the project and thereafter as long as required.

8.3 NLR

NLR will not be using any other national or departmental procedures for data management.

8.4 KTH

KTH is not using any national or departmental procedures for data management.

8.5 TUE

TUE is not using any national or departmental procedures for data management.

8.6 SISW

SISW is not using any national or departmental procedures for data management.